

Weekly Progress Report

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Domain: Internet of Things (IoT) / Electronics

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Week Ending: 02

I. Overview:

This week, the primary focus was on designing the logic and program flow for the new Automatic Door Control System, specifically integrating motion sensing with motor control.

II. Achievements:

1. USC_TIA Familiarization:

- Explored sensor integration concepts, focusing on how to interface digital input sensors with motor driver modules on the development board.

2. Project Contributions:

- Name of the project: Automatic Door Control System
- Designed the core program flow, beginning with declaring the necessary variables for the motor controller (Motor_pin 1, Motor_pin 2).
- Implemented conditional logic to evaluate the incoming values from the PIR Sensor.
- Configured the output states: programmed the system to close the door when Motor_pin 1 is High and Motor_pin 2 is Low, and to open the door when Motor_pin 1 is Low and Motor_pin 2 is High.

3. Learning Python & IoT Programming:

- Learned the fundamental logic of H-bridge motor drivers and how microcontroller pins dictate the direction of DC motors based on sensor triggers.

III. Challenges:

1. USC_TIA Integration:

- Encountered minor challenges in mapping the theoretical flowchart logic into the specific syntax and pin configurations required by the board.

2. Project Complexity:

- Faced complexity in determining the optimal timing and delay logic after the PIR sensor detects motion to ensure the door remains open for a safe and practical duration before closing.

IV. Learning Resources:

1. USC_TIA Documentation:

- Utilized schematics and component documentation to understand the behavior of the PIR sensor and the expected output for the motor pins.

2. IoT Learning Resources:

- Consulted online tutorials regarding basic motor driver circuitry and motion detection algorithms to refine the system's flow.

V. Next Week's Goals:

1. USC_TIA Enhancement:

- Deploy and test the implemented code directly on the physical hardware to observe real-world sensor accuracy and motor response.

2. Project Development:

- Refine the control code to include safety mechanisms, such as interrupt routines to keep the door open if continuous motion is detected while closing.

VI. Additional Comments:

Designing the visual program flow before jumping into the code was a highly effective step this week. It made the conditional logic for the motor states much clearer and streamlined the development process.